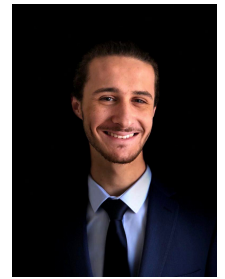


Jozef Sawan

Mechanical Engineer — Simulation, Design Office & R&D



Saint-Germain-en-Laye, France • +33 6 52 62 95 83 • jozefsawan@protonmail.com

Portfolio: jozefsawan.github.io/en • GitHub: jozefsawan • Nationality: French / Austrian

Mechanical & Materials Engineer with a versatile profile and a strong problem solving mindset. Solid background in design, engineering analysis, materials and industrial methods, with experience in R&D and technical tool automation. Quick to adapt to business needs and able to contribute to product and process performance improvements.

Skills

Structural Mechanics

Strength of materials, sizing and dimensioning, stress and strain, structural dynamics (plates and beams).

Materials

Mechanical characterization, material selection, elastic and plastic behavior, fatigue basics, selection criteria.

Simulation & Engineering Analysis

FEM 1D/2D/3D, meshing, convergence, boundary conditions, numerical optimization, model/test validation.

Vibration & Acoustics

Modal analysis, frequency/time response, vibration propagation, test vs simulation correlation.

Manufacturing / Industrial Methods

Manufacturing processes, quality basics, continuous improvement, technical documentation.

Programming

Python (automation, post-processing), MATLAB.

Tools

- Simulation: COMSOL, Abaqus, ANSYS
- CAD: SolidWorks
- Materials: CES EduPack
- Computing / Docs: Python, MATLAB, LaTeX, Office

Languages

- French — Native
- English — C2
- German — A2 (Goethe)

Strengths

- Analytical mindset and scientific rigor
- Autonomy, clear technical documentation
- Project culture (entrepreneurial experience)
- Curiosity and fast learning

Interests

Tennis, badminton, hiking
3D printing, CAD/Blender

Experience

R&D Internship — Vibration & Acoustics

Feb 2025 – Aug 2025

VENATHEC

- FEM modelling of structures in COMSOL (vibration propagation)
- Modal analysis and multi-source simulations (loading scenarios)
- Model-to-test correlation: simulation/measurement validation
- Python scripts: results extraction, comparison, automated figures
- Method improvements: fewer iterations and reduced computation time

Software & Automation Internship

Dec 2022 – Sep 2023

iGoFlex

- Embedded application: integration, testing and iterations
- Automated integration of customer orders into a database
- Technical revamp of the e-commerce website (maintenance, performance)

Software Internship

Jul 2022 – Aug 2022

Phonexpat

- Development of an automated modem configuration tool
- English-speaking technical support and installation assistance

Entrepreneurship — Pépité Program (SNEE)

Mar 2024 – Jul 2024

JoCard

- Project management, product design, user iterations
- Communication, organization and prioritization in real conditions
- Marketing and creation/management of the e-commerce website

Education

M.Sc. — Mechanical Engineering & Materials (GMM)

2023 – 2025

Université Bretagne Sud

B.Sc. — Mechanical Engineering & Industrial Engineering (GMP)

2018 – 2023

Université Grenoble Alpes

Master Project

- Sheet metal deep drawing with anisotropy: modelling, stress analysis, and impact of material properties on formability